

Files in Python

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What are Files?

- ▶ A **file** in Python is a collection of data stored on a computer that can be **read, written, or modified** using Python's built-in file-handling functions.
- ▶ It allows for **permanent storage of data** beyond the execution of a program.
- ▶ Allows storing and manipulating data permanently.
- ▶ Python provides built-in functions to read, write, and modify files.
- ▶ Common uses: Logging, data storage, etc.

Opening a File

► Syntax:

```
file = open("filename.txt", "mode")
```

► Modes:

- 'r' - Read mode (default)
- 'w' - Write mode
- 'a' - Append mode
- 'x' - Exclusive creation mode
- 'b' - Binary mode

Reading a File

- ▶ Opening a file:

```
file = open("example.txt", "r")  
print(file.read()) # Reads entire file  
file.close()
```

- ▶ Other methods:

- `file.readline()` - Reads one line
- `file.readlines()` - Returns list of lines

Writing to a File

```
file = open("example.txt", "w")
```

```
file.write("Hello, World!\n")
```

```
file.close()
```

- 'w' erases existing content.
- Use 'a' mode to append instead.

Closing a File

- ▶ Always close a file to free resources.
- ▶ Use `file.close()`
- ▶ or `with` statement:
 with open("example.txt", "r") as file:
 data = file.read()

Using `with` for File Handling

- ▶ Ensures the file is properly closed.
with open("data.txt", "w") as file:
 file.write("Using with statement")

Removing a File

- ▶ Use `os` module:

```
import os  
os.remove("example.txt")
```

- ▶ Check if file exists before removing:

```
if os.path.exists("example.txt"):  
    os.remove("example.txt")
```


Programs

- ▶ Write a program to read the text from a given file 'myfile.txt' and find out whether it contains the word 'python'.
- ▶ Write a program to generate multiplication table of 5 and store in a file namely Multiplication.txt
- ▶ Write a program to make a copy of a text file "name.txt"
- ▶ Write a program to find out whether a file is identical & matches the content of another file.
- ▶ Write a program to delete out the content of a file using python
- ▶ Write a program to delete a file using python